# TEEN MENTAL HEALTH & SOCIAL MEDIA ANALYTICS

## 1. Project Overview

Developed an interactive analytics dashboard using Python and Power BI to analyze social media usage among teenagers and its impact on mental health. The project helps identify patterns related to anxiety, depression, sleep quality, and self-esteem, supporting better decision-making through data visualization and business insights.

### Project Objectives

- Analyze social media usage among teenagers.
- Identify mental health patterns.
- Study the relationship between screen time and anxiety.
- Evaluate sleep quality and self-esteem.
- Create an interactive Power BI dashboard.
- Provide actionable recommendations.

## 2. Data Source

### Source Description

Dataset obtained from Kaggle.

### Domain

Healthcare Analytics / Social Media Analytics

### Dataset Type

CSV Dataset

## 3. Problem Statement

Analyzing teenage social media usage and mental health data by cleaning, transforming, and visualizing datasets using Python and Power BI to identify usage trends, anxiety levels, depression patterns, sleep quality issues, and self-esteem variations.

## 4. Attribute (Column / Feature) Details

| Attribute | Data Type | Description |
|---|---|---|
| Age | Numeric | Age of teenager |
| Gender | Text | Gender of participant |
| Usage_Hours | Numeric | Daily social media usage |

| | | |
|---|---|---|
| Anxiety_Level | Numeric | Anxiety score |
| Depression | Numeric | Depression score |
| Sleep_Quality | Text | Sleep quality level |
| Self_Esteem | Numeric | Self-esteem score |
| Academic_Performance | Numeric | Academic score |
| Usage_Category | Text | Low, Medium, High usage |

## 5. Tools & Technologies

### Python

- Pandas
- NumPy
- Matplotlib
- Seaborn

### Power BI

- Data Modelling
- DAX Calculations
- Dashboard Creation
- Interactive Visualizations

## 6. Data Pre-Processing

### Tasks Performed

- Removed duplicate records.
- Handled missing values.
- Standardized column names.
- Checked data types.
- Created Usage Category column.
- Created High Usage indicator.
- Performed outlier detection.
- Saved cleaned dataset.

## 7. Data Modelling and DAX (Power BI)

### Data Model

- Imported cleaned CSV into Power BI.
- Created relationships where required.
- Built calculated columns and measures.

## DAX Measures

```
Total Teenagers = COUNTROWS('TeenMentalHealth')

Average Usage Hours = AVERAGE('TeenMentalHealth'[Usage_Hours])

Average Anxiety Level = AVERAGE('TeenMentalHealth'[Anxiety_Level])

Average Depression = AVERAGE('TeenMentalHealth'[Depression])
```

## Screenshot of Data Model

(Paste Screenshot Here)

# 8. Analysis and Visualizations (Power BI)

## Dashboard Features

### *KPI Cards*

- Total Teenagers
- Average Usage Hours
- Average Anxiety Level
- Average Depression Score

### *Bar Chart*

Gender vs Anxiety Level

### *Pie Chart*

Usage Category Distribution

### *Line Chart*

Age vs Average Usage Hours

### *Heatmap*

Correlation between:

- Usage Hours
- Anxiety Level
- Depression
- Self-Esteem

### *Slicers*

- Gender
- Age
- Usage Category
- Sleep Quality

### Screenshot of Dashboard

(Paste Power BI Dashboard Screenshot Here)

## 9. Insights & Conclusions

### Descriptive Analysis

- Most teenagers spend between 2–5 hours daily on social media.
- High social media usage is common among teenagers.
- Anxiety and depression levels increase with usage hours.
- Poor sleep quality is observed among heavy users.
- Self-esteem tends to decrease with higher screen time.

### Diagnostic Analysis

- Excessive social media usage contributes to higher anxiety levels.
- Poor sleep quality is linked with increased screen time.
- Teenagers with low self-esteem show higher mental health risks.
- High social media engagement may negatively affect emotional well-being.

### Predictive Analysis

- Teenagers with high screen time are more likely to experience anxiety.
- Mental health risks may increase if excessive usage continues.
- Poor sleep quality may become more common among heavy users.

### Prescriptive Analysis

- Encourage healthy screen-time habits.
- Promote digital wellness awareness.
- Support mental health programs in schools.
- Encourage outdoor and physical activities.
- Educate teenagers about responsible social media usage.

## 10. Conclusion

This project successfully analyzed the impact of social media usage on teenage mental health using Python and Power BI.

The dashboard provided meaningful insights into:

- Social Media Usage Patterns
- Anxiety Levels
- Depression Trends

- Sleep Quality
- Self-Esteem

Through interactive visualizations and analytical techniques, the project demonstrated how data analytics can help identify mental health risks and support informed decision-making. The findings highlight the importance of balanced social media usage and mental health awareness among teenagers.